

NGUYEN KHAC BAO

AI Engineer Intern – Computer Vision Dataset & QA

Ho Chi Minh City, Vietnam | nguyenkhacbaowork564@gmail.com | (+84) 039 586 4429 | github.com/NguyenKhacBao564

CAREER OBJECTIVE

AI Engineer Intern candidate focused on **computer vision dataset quality**, annotation workflows, metadata tracking, QA reports, and **real-world edge-case analysis**. Hands-on with image/video datasets, YOLO/VOC formats, keypoint annotation, Python validation scripts, and model-aware dataset improvement.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

2022 – 2027

Bachelor of Information Technology

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

TECHNICAL SKILLS

Data & Annotation: YOLO/VOC labeling, keypoint annotation, metadata tracking, train/val/test split, dataset versioning, QA reports

Computer Vision: Object detection, pose estimation, image classification, OCR, OpenCV, YOLOv8n, YOLO-Pose

Python / Tools: Python, Pandas, Streamlit, PyYAML, JSON/CSV processing, Git/GitHub, Linux, Docker

Dataset QA: class distribution checks, duplicate filtering, malformed label detection, blur/dark flags, occlusion and edge-case analysis

LANGUAGES

Vietnamese: Native | English: TOEIC 610/990; technical reading proficiency for documentation and engineering articles.

PROJECTS

FallGuard Labeling Tool – Computer Vision Dataset Assistant

2026

Personal Project | Python, Streamlit, OpenCV, YOLO-Pose, Pandas | Source: [GitHub](#)

- Built an end-to-end **computer vision dataset assistant** for fall-detection data: image/video ingestion, **metadata tracking**, auto pose-label generation, human review, QA reports, and reviewed-only export.
- Developed a **Streamlit review UI** with bounding boxes, **13 keypoints**, confidence scores, label correction, uncertain/exclude actions, and manual keypoint editing.
- Processed 5 surveillance videos into **656 extracted frames**; produced **332 reviewed exportable samples** and tracked excluded samples, keypoint CSVs, event logs, QA reports, and train/val/test split.
- Added evaluation reports for **auto-label accuracy**, correction rate, pose detected/missing rate, and **90.2% coarse fall/non-fall agreement** to expose noisy real-world CCTV data issues.

Real-Time Face Mask Compliance Detection System

2026

Personal Project | Python, YOLOv8n, OpenCV, FastAPI, Docker, Vast.ai | Source: [GitHub](#)

- Converted PWMFD from **VOC/XML to YOLO format** with **9,205 images** and **18,528 labeled face instances**; handled class mapping, split validation, and empty-label cases.
- Generated **dataset audit reports** for class distribution, malformed labels, missing files, and sample visualization before model training.
- Trained a YOLOv8n 3-class detector for correct_mask / incorrect_mask / no_mask, achieving **0.970 mAP@50** and **0.749 mAP@50:95**.
- Collected real-world test metadata such as lighting, occlusion, blur, reflection, and mask subtype for structured **edge-case analysis**.

Traffic Violation Detection & License Plate Recognition

2026

Personal Project | Python, YOLO, OpenCV, ByteTrack, OCR | Source: [GitHub](#)

- Built a fixed-camera video pipeline for vehicle detection/tracking, traffic-light ROI state, stop-line crossing logic, violation events, **plate OCR**, and evidence generation.
- Prepared a compact CCPD subset of **8,598 images** with YOLO plate labels, OCR crops, train/val/test splits, negative samples, and manifest metadata.
- Created audit artifacts to flag missing files, **invalid YOLO boxes**, duplicates, dark/blur/small-plate samples, and **contact sheets** for human QA.

ACTIVITIES

AI Research Group – University Club

2025 – Present

Member

- Discussed Machine Learning, Computer Vision, **dataset quality**, data augmentation, and object detection topics.